Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/10/2004
PI Name:	Miller, Jack Ph.D.		
Project Title:	Radiation Transport Properties of Potential In Situ-Developed Regolith-Epoxy Materials for Martian Habitats		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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Task Description:	The composites will be prepared by NASA Langley Research Center using simulated Martian regolith. Initial evaluation of the radiation shielding properties will be made using transport models developed at NASA-LaRC, and the results of these calculations will be used to select the composites with the most favorable radiation transmission properties. These candidate materials will then be empirically evaluated at particle accelerators which produce beams of protons and heavy charged particles comparable in energy to the radiation at the surface of Mars. To evaluate the radiation transport properties of epoxy-Martian regolith composites with respect to the protons and heavy charged particles in the galactic cosmic rays (GCR) which bombard the Martian surface. Such composites, which would use both in situ materials and chemicals fabricated from elements found in the Martian atmosphere, are candidates for use in habitats on Mars. The secondary objective is to evaluate fabrication methods which could lead to technologies for in situ fabrication. Knowledge of the radiation transmission properties of Martian soil, or regolith, is potentially critical for the planning of manned exploratory missions to Mars. Most, if not all scenarios for such missions assume that crew members will spend weeks or months on the planet's surface. During this time they will be exposed to charged subatomic particle radiation to a substantially greater degree than on the surface of the earth, which is shielded by a much stronger magnetic field and a much thicker atmosphere. Since in order to minimize the amount of material that will need to be transported to Mars it will be imperative to make as much use as possible of Martian resources, composite materials which exploit the radiation shielding properties and plentiful availability of Martian regolith would be favored for habitat construction. While no exploratory missions are currently scheduled, the long lead time required to mount these missions dictates that as much information as possible be gathered in advance, so that it can be in place when the decision to travel to Mars is made. This is especially true for issues bearing on radiation exposure and protection, which have the potential to influence every stage of the mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Model calculations and selected experiments at particle accelerators were used to evaluate the radiation transmission properties of regolith composites. Several regolith composites were modeled using two different benchmarks of biological effectiveness. The composites were then fabricated at NASA-LaRC and were exposed to several different accelerated HZE beams at the BNL AGS, NSRL and NIRS HIMAC. Charged fragment and neutron spectra were recorded for nuclear fragments produced in the forward direction. The accelerator-based measurements produced detailed data on the radiation transport properties of selected materials with respect to significant components of the GCR HZE. These data were used both to quantify the radiation shielding effectiveness of the candidate materials and to improve the accuracy of the models. Ultimately the models, in combination with data from radiobiological experiments now in progress, will comprise a design tool for use by mission planners, flight surgeons and radiation health specialists.
Bibliography Type:	Description: (Last Updated: 01/21/2020)